17.

Average

Type - 7

1. Find the average of 3/4, 5/8, 7/12, 15/16

(a) 139/192 (b) 135/64
(c) 11/32 (d) 21/64

RRB RPF SI – 05/01/2019 (Shift-II)

Ans : (a)

$$\text{Average} = \frac{\text{sum of terms}}{\text{number of terms}}$$

$$\text{Average} = \frac{\frac{3}{4}+\frac{5}{8}+\frac{7}{12}+\frac{15}{16}}{4}$$

$$\text{Average} = \frac{\frac{36+30+28+45}{48}}{4}$$

$$= \frac{139}{48\times 4} = \frac{139}{192}$$

$$\text{Average} = \frac{139}{192}$$

2. The average of 5 consecutive even numbers is 40. Find the value of smallest number among these numbers.

(a) 35 (b) 36
(c) 44 (d) 48

RRB RPF Constable – 19/01/2019 (Shift-II)

Ans : (b) Let the 5 consecutive even numbers are x, x +2, x + 4, x + 6 and x + 8

According to question,

$$\frac{x+x+2+x+4+x+6+x+8}{5} = 40$$

5x + 20 = 200

5x = 180

$\boxed{x = 36}$

Therefore, smallest number is x = 36.

3. If the average of 4 numbers a, b, c, d is 26. If the average of 'a' and 'b' is 19.5 then average of c and d will be.

(a) 33 (b) 35.5
(c) 31.5 (d) 32.5

RRB RPF Constable – 18/01/2019 (Shift-I)

Ans : (d) According to first condition,

$$\frac{a+b+c+d}{4} = 26$$

∴ a + b + c + d = 104 ....(i)

According to second condition,

Average of a and b = 19.5

a + b = 39 ....(ii)

putting the value of a+b of equation (ii) in equation (i)

∵ a + b + c + d = 104

39 + c + d = 104

c + d = 104 – 39

c + d = 65

Therefore Average of c and d $= \frac{c+d}{2}$

$= \frac{65}{2} = 32.5$

4. W, X, Y and Z are four numbers arranged in ascending order.The average of smallest 3 numbers is 22 while the average the largest 3 numbers is 28. Find the range of data?

(a) 19 (b) 18
(c) 17 (d) 16

RRB Group-D – 10/10/2018 (Shift-II)

Ans : (b) Let the numbers are w < x < y < z

w + x + y = 22 × 3 ____ (i)

x + y + z = 28 × 3 _____ (ii)

Range = Greatest number− smallest number

by subtracting equation (i) from equation (ii)

⇒ z – w = 84 – 66 = 18

5. The average of three numbers is 7. The average of first two is 5, while the average of last two is 8. Which are three consecutive numbers?

(a) 3, 7 and 9 (b) 2, 8 and 8
(c) 5, 5 and 11 (d) 4, 6 and 10

RRB Group-D – 15/10/2018 (Shift-I)

Ans : (c) Let the numbers are *x, y, z*

According to question–

⇒ $\frac{x+y+z}{3} = 7$ -----(i)

⇒ $\frac{x+y}{2} = 5$ -----(ii)

⇒ $\frac{y+z}{2} = 8$ -----(iii)

⇒ from equation (i) and (ii)

⇒ z = 21-10 = 11, z = 11

⇒ putting the value of z in equation (iii)

⇒ y = 16-11 = 5, y = 5

⇒ putting the value of y in equation (ii)

⇒ x = 10-5 = 5, x = 5

Therefore, the numbers are 5,5 and 11

6. Arranging the four digits in ascending order, their order is w, x, y and z. The average of smallest three digit is 25.5. While the largest three digit average is 29.5. Find the rang of data.

(a) 13 (b) 12
(c) 10 (d) 11

RRB Group-D – 24/09/2018 (Shift-I)

Ans : (b) Let the number are z > x > y > w
According to the question,

$\frac{w+x+y}{3} = 25.5$

w + x + y = 76.5 .....(1)

$\frac{x+y+z}{3} = 29.5$

x + y + z = 88.5 .....(2)

by subtracting equation (i) from equation (ii)

$\boxed{z - w = 12}$

Therefore, range = greatest digit – smallest digit

z – w = 12

7. The average of three numbers is 8. The average of first two numbers is 6 and the average of last two numbers is 9. Find the three numbers.

(a) 4, 9, 9 (b) 4, 8, 12
(c) 5, 7, 12 (d) 6, 6, 12

RRB Paramedical Exam – 20/07/2018 (Shift-III)

Ans : (d) Let the three numbers are a, b and c.
According to question,

$\frac{a+b+c}{3} = 8$

a + b + c = 24 ------(i)

$\frac{a+b}{2} = 6$

a + b = 12 ------(ii)

and $\frac{b+c}{2} = 9$

b + c = 18 ------(iii)

from equation (i) , (ii) and (iii)
a = 6, b = 6 and c = 12
∴ c = 24 - 12 = 12
Therefore, three numbers are 6, 6 12 respectively.

8. The average of three number is 28. If in the smallest number 2 is added and from the largest number 5 is subtracted, then the middle digit becomes the arithmetic mean, while the range of this new set becomes 36. What is the largest number among these three numbers of the original set.

(a) 50 (b) 48
(c) 47 (d) 45

RRB Group-D – 01/12/2018 (Shift-II)

Ans : (a) Let all the three numbers are x, y, z and x < y < z

x + y + z = 28 × 3
x + y + z = 84 --------(i)

According to condition-

$\frac{(x+2)+y+(z-5)}{3} = y$

x + y + z = 3y + 3
x – 2y + z = 3 --------(ii)

with the same condition range is

z – 5 – (x + 2) = 36
z – x = 43 --------(iii)

by subtracting equation (ii) from equation (i)

y = 27

from equation (i)

x + y + z = 84
x + z = 84 – 27
x + z = 57
+ z – x = 43
2z = 100
z = 50

There, largest number of original set = 50

9. The average of three numbers is 28. If 7 is added in the smallest number and 10 is subtracted from the largest number, then the middle digit is the arithmetic mean and the range of this new set becomes 20. What is the largest number among the original set of three numbers?

(a) 47 (b) 40
(c) 45 (d) 50

RRB Group-D – 24/10/2018 (Shift-III)

Ans. (a) : Let the first, second and third number is x, y and z respectively and x < y < z
According to question,
sum of all three numbers = 28 × 3
x + y + z = 84 ....(i)
according to second condition

$\because \frac{z-10+y+x+7}{3} = y$

x +y+ z – 3 = 3y
x + z – 2y = 3 .....(ii)
z – 10 – x – 7 = 20
z – x = 37 .....(iii)
from equation (i), (ii) and (iii)
x + y + z = 84

$z - 37 + \frac{2z-40}{2} + z = 84$

2z - 74 + 2z – 40 + 2z = 168
6z = 168 + 114
6z = 282
z = 47

10. Four numbers a, b, c and d are such that their total average 39. The average of a and b is 29.5. The average of c and d will be –

(a) 48.5 (b) 48
(c) 49.5 (d) 47.5

RRB Group-D – 09/10/2018 (Shift-II)

Ans. (a) : sum of the number = a+b+c+d = 39×4
= 156
sum of number = a+b = 29.5 × 2 = 59
sum of the number = c+d = [(a+b+c+d) – (a+b)]
= [156 – 59] = 97

∴ Average of number = c + d $= \frac{97}{2} = 48.5$

11. Average of first 50 natural number is–

(a) 26.5 (b) 25.5
(c) 26 (d) 25

RRB NTPC 17.01.2017 Shift-3

Ans : (b) ∵ Average of first n natural numbers

$$= \frac{n+1}{2}$$

∴ Average of first 50 natural numbers $= \frac{50+1}{2} = 25.5$

12. The sum of 7 numbers is 1050. The average of first 3 numbers is 120, fourth number is 126, then find the average of last 3 numbers.

(a) 200 (b) 165
(c) 188 (d) 173

RRB NTPC 05.04.2016 Shift : 2

Ans : (c) Sum of first three numbers = 120×3 = 360
Sum of last three numbers = 1050– (360+126)
= 1050–486
= 564

Average of last three numbers are $= \frac{564}{3} = 188$

13. Find the averge of 8, 5, 6, 3, 7, 4, 3, 9

(a) 5.63 (b) 5.64
(c) 5.65 (d) 5.66

RRB NTPC 05.04.2016 Shift : 2

Ans : (a) Average $= \frac{8+5+6+3+7+4+3+9}{8}$

$$= \frac{45}{8} = 5.625 = 5.63$$

14. Find the averge of 1, 9, 7, 3, 5, 5, 6, 4, 2, 8

(a) 3 (b) 4
(c) 5 (d) 6

RRB NTPC 05.04.2016 Shift : 3

Ans :(c) Average $= \frac{1+9+7+3+5+5+6+4+2+8}{10}$

$$= \frac{50}{10} = 5$$

15. Find the average of first 20 multiples of 7–

(a) 66.5 (b) 67.5
(c) 73.5 (d) 74.5

RRB NTPC 03.04.2016 Shift : 3

Ans : (c) Sum of the first n coefficients of K

$$= \frac{K[n(n+1)]}{2}$$

Sum of the first 20 coefficients of 7 $= \frac{7\times[20(20+1)]}{2}$

$$= \frac{7\times20\times21}{2}$$

= 1470

$$\boxed{\text{Average} = \frac{\text{sum total}}{n}}$$

Average $= \frac{1470}{20}$

Average = 73.5

16. Find the average of first 20 multiples of 8–

(a) 78 (b) 80
(c) 84 (d) 82

RRB NTPC 02.04.2016 Shift : 1

Ans : (c) 8, 16, 24 …… 160

Sum of terms = $\frac{n}{2}(a+\ell)$

$$= \frac{20}{2}(8+160) = 10 \times 168 = 1680$$

∴ Average $= \frac{1680}{20} = 84$

17. What is the average of first 30 multiples of 9?

(a) 142 (b) 138.5
(c) 139.5 (d) 143.5

RRB NTPC 16.04.2016 Shift : 1

Ans : (c) Sum of the first n coefficient of K $= \frac{K[N(N+1)]}{2}$

∴Sum of the first 30 coefficient of 9 $= \frac{9[30(30+1)]}{2}$

$$= \frac{9\times30\times31}{2}$$

= 4185

Average $= \frac{4185}{30} = 139.5$

18. Three numbers are given in which second number is thrice the first, and twice of the third number. If the average of three numbers is 66. Then find the first number?

(a) 36 (b) 54
(c) 108 (d) 72

RRB NTPC 12.04.2016 Shift : 3

Ans : (a) Let the second number = x

∴ first number $= \frac{x}{3}$

third number $= \frac{x}{2}$

∴ from question,

$$\frac{\frac{x}{3}+x+\frac{x}{2}}{3} = 66$$

$$\Rightarrow \frac{2x+6x+3x}{6\times3} = 66$$

$$\Rightarrow \frac{11x}{18} = 66$$

$$\Rightarrow x = 6 \times 18 = 108$$

∴ first number $= \left(\frac{x}{3}\right) = \frac{108}{3} = 36$

19. The average of 5 consecutive numbers is 50. What is the difference of the products of the largest and smallest number to the product of fourth and second number?

(a) 3 (b) –3
(c) 0 (d) 10

RRB NTPC 27.04.2016 Shift : 1

Ans : (b) Let the 5 consecutive numbers are x, x + 1, x + 2, x + 3 and x + 4

$$\therefore \quad \frac{x+(x+1)+(x+2)+(x+3)+(x+4)}{5}=50$$

$5x + 10 = 250$

$5x = 240$

$x = 48$

$\therefore$ from question,

$x(x + 4) - (x + 1)(x + 3)$

$= x^2 + 4x - (x^2 + 4x + 3) = -3$

20. The average of 5 consecutive numbers is 10 then which will be the middle number?

(a) 10 (b) 11

(c) 8 (d) 9

RRB NTPC 29.04.2016 Shift : 3

Ans. (a) : Let the 5 consecutive numbers are x, x + 1, x + 2, x + 3 and x + 4

Total sum –

$x + (x + 1) + (x + 2) + (x + 3) + (x + 4) = 10 \times 5$

$5x + 10 = 50$

$5x = 40$

$x = 8$

$\therefore$ Middle number $= (x + 2) = 8 + 2 = 10$

21. The average of 5 consecutive number is 100, find the first number?

(a) 98 (b) 99

(c) 100 (d) 101

RRB NTPC 30.04.2016 Shift : 1

Ans : (a) Let the five consecutive numbers are

$x, x+1, x+2, x+3, x+4$

$$\therefore \quad \frac{x+x+1+x+2+x+3+x+4}{5}=100$$

$5x+10=500$

$5x=490$

$x=98$

Therefore, first number is 98.

22. The average of 5 consecutive number is 100, then the difference of the square of highest number and the square of the lowest number will be?

(a) 800 (b) 990 (c) 900 (d) 1000

RRB NTPC 30.04.2016 Shift : 2

Ans : (a) Let the first consecutive numbers are: x, x +1, x + 2, x + 3 and x + 4 respectively

According to the question,

$$\frac{x+x+1+x+2+x+3+x+4}{5}=100$$

$$\frac{5x+10}{5}=100$$

$5x + 10 = 500$

$5x = 490$

$\boxed{x=98}$

Largest number $= x + 4 = 98 + 4 = 102$

Smallest number $= x = 98$

Therefore, difference between squares of largest and smallest numbers

$=(102)^2-(98)^2=(102+98)(102-98)=200\times 4=800$

Type - 7

23. The average mark of 3 students of a class in a test is 18 out of 25. Two new students take test. How much is such minimum number that can be obtained by the new student and this be less than other students and the total average of other 5 students is raised to 20.

(a) 23 (b) 20

(c) 21 (d) 22

RRB RPF SI – 06/01/2019 (Shift-II)

Ans : (c) Maximum marks of three students

$= 25 \times 3 = 75$ Marks

and sum of their marks $= 18 \times 3 = 54$ marks

Total marks of 5 students $= 20 \times 5 = 100$ marks

5 – 3 = 2 marks obtained by students $= 100 - 54 = 46$

46

(because maximum marks is 25) 25 21

Maximum marks minimum marks

There minimum 21 marks can be obtained by a student.

24. The number of students of three groups of a college G_1, G_2 and G_3 is 20, 40 and 60 respectively. The average marks obtained by group G_1, G_2 and G_3 is 50%, 60% and 70% what is the average marks of all the sudents of the college?

(a) 62% (b) 61%

(c) 60% (d) 63%

RRB Group-D – 03/10/2018 (Shift-II)

Ans : (d) According to question-

Total marks of $G_1 = 20 \times 50 = 1000$

Total marks of $G_2 = 40 \times 60 = 2400$

Total marks of $G_3 = 60 \times 70 = 4200$

Total marks of G_1, G_2 and G_3

$= 1000 + 2400 + 4200 = 7600$

Total students $= 20 + 40 + 60 = 120$

Average of marks of all students $=\frac{7600}{120}$

$= 63.3\% \approx 63\%$

25. The average marks obtained by Reena in 16 tests is 26. The average marks obtained by shreya till now is 24, but she has taken only 12 tests till now. How much average marks has to be obtained by shreya in order to equalise the performance of Reena in the rest tests.

(a) 28 (b) 32

(c) 30 (d) 26

RRB Group-D – 06/12/2018 (Shift-III)

Ans. (b) : Average of marks obtained by Reena in 16 examinations = 26

$\therefore$ Total marks $= 26 \times 16 = 416$

Average of marks obtained by Shreya in 12 examinations = 24

$\therefore$ Total marks $= 24 \times 12 = 288$

Total difference of Reena and Shreya's marks $= 416 - 288 = 128$

Difference between their examinations = 16 – 12 = 4

Average score of 4 tests of Shreya = $\frac{128}{4}$ = 32 marks

Therefore, In order to perform like reena, shreya has to score 32 marks in 4 examinations.

26. In a class there were 9 boys and some girls. In a test, the mean score obtained by the boys was 13 while that obtained by the girls was 15. If the overall average was 14.28, what was the total number of students in the class?

(a) 24 (b) 25
(c) 26 (d) 27

RRB Group-D – 18/09/2018 (Shift-I)

Ans. (b) : Let the number of Girls = x

According to question -

$9 \times 13 + 15 \times x = 14.28\ (x+9)$

$117 + 15x = 14.28x + 128.52$

$0.72x = 11.52$

$x = \frac{1152}{72}$

$x = 16$

Therefore number of girls = 16

Number of total students = x + 9 = 16 + 9 = 25

27. In a 25 marks test the mean of marks obtained by three students is 16. Two new students took part in the test. In order to raise the average of students to 19 how much marks have to be obtained by the new student who obtains less marks than other new student?

(a) 22 (b) 21
(c) 20 (d) 23

RRB Group-D – 01/11/2018 (Shift-II)

Ans : (a) Let the three boys are x, y and z

According to question,

$x + y + z = 48$ ----- (i)

Let there be tow newly included students A and B

Then sum of the marks obtained by 5 students =

$x + y + z + A + B = 95$ -----(ii)

from equation (i) and (ii),

$A + B = 95 - 48$

$A + B = 47$ -----(iii)

Maximum marks for A + B = 25 + 25 = 50 marks

If we give the highest marks to A, then the minimum marks obtained by B is 47 – 25 = 22 marks.

28. The average marks obtained in a test of 18 boys of a class is 16, while the average of the total 30 students of the class is 18.1 what is the average of the girls?

(a) 21.25 (b) 20.5
(c) 20.75 (d) 21

RRB Group-D – 25/09/2018 (Shift-II)

Ans : (a) Average of marks obtained by 18 boys in the test = 16

total marks = 16 × 18 = 288

Average of all 30 students = 18.1

total marks = 18.1 × 30 = 543

total obtained mark by 12 girls = 543 – 288

= 255

Average of all 12 girls = $\frac{255}{12}$ = 21.25

29. The average of the marks of three students in an examination with a total of 45 marks is 38. Two new students participated in the examination. What is the lowest number of marks that can be scored by a new student who has scored less than another new student, making the total Average mark of five students to be 40?

(a) 41 (b) 42
(c) 40 (d) 43

RRB Group-D – 19/09/2018 (Shift-II)

Ans. (a) Sum of marks of all 5 students = 40 × 5 = 200

Sum of mark of first three students = 38 × 3 = 114

Sum of mark of remaining two new students = 200–114 = 86

Maximum marks in examination = 45

Hence, one of both new students can score a maximum 45 marks.

and second student will be able to get minimum 86–45 = 41 marks

30. The average marks of 12 test obtained by Raghuveer is 25. The average of Rumela is 23 marks till now but she had taken part only in 8 tests. How much marks Rumela has to get in the rest 4 test in order to equal Raghuveer's average?

(a) 27 (b) 29
(c) 26 (d) 28

RRB Group-D – 22/09/2018 (Shift-I)

Ans : (b) Total score in 12 tests by Raghuveer = 12 × 25 = 300

Total score in 8 tests by Rumela = 8 × 23 = 184

Required marks of 4 tests to reach at the level of Raghuveer = 300 – 184 = 116

Intended Average = $\frac{116}{4}$ = 29

31. The group of five students took an exam. One more students later joined the group after taking exam. If his marks are included then the average of group is raised by 2 marks. Without including his average marks this student scored the ___ more marks?

(a) 18 (b) 14
(c) 12 (d) 15

RRB Group-D – 26/09/2018 (Shift-I)

Ans : (c)

Let the five students are A, B, C, D, E and their average = x

According to question,

$\Rightarrow A + B + C + D + E = 5x$ _____(i)

Let the newly included student = F

$\Rightarrow$ A+ B+ C + D + E + F = 6 (x+2) = 6x + 12 ___ (ii)
from equation (i) and (ii)

$\Rightarrow 6x + 12 - 5x = F$

$\Rightarrow F = x + 12$

Hence, the marks of the newly included student is 12 more than the average marks.

32. There were 28 boys and some girls in a class. In an exam the average marks obtained by boys was 12.5. While the girls obtained 14.5 if the total average was 13.1 then what was the total no. of students in the class?

(a) 42 (b) 40
(c) 44 (d) 38

RRB Group-D – 03/10/2018 (Shift-I)

Ans : (b) Let the number of girls = x
According to question,

$$13.1 = \frac{28 \times 12.5 + x \times 14.5}{28 + x}$$

366.8 + 13.1x = 350 + 14.5 x
1.4 x = 16.8
x= 12
number of boys in class = 28
number of girls in class = 12
Total number = 28 + 12 = 40

33. In a class of 40 students the ratio of boys and girls is 7:3. Average marks of boys is 65 and the average marks of the girls is 72. What is the average mark of the whole class?

(a) 67.1 (b) 68.4
(c) 68.3 (d) 68.2

RRB Group-D – 31/10/2018 (Shift-II)

Ans : (a) No. of boys = $40 \times \frac{7}{10} = 28$

No. of girls = $40 \times \frac{3}{10} = 12$

Ratio of Boys and Girls = 7 : 3
Total marks of boys = 65×7 = 455
Total marks of girls = 72× 3 = 216
Total marks = 671
Number = 10

Therefore, Average of whole class $= \frac{671}{10} = \boxed{67.1}$

34. In a class of 45 students. The ratio of boys and girls is 4:5 average marks of boys is 75 and that of girls is 82. What is the approximate average marks of the class?

(a) 78.6 (b) 78.5
(c) 78.9 (d) 79.0

RRB Group-D – 05/11/2018 (Shift-I)

Ans. (c) : Let the number of boys and girls in a class of 45 students are 4x, 5x respectively.

$\therefore$ 4x + 5x = 45
9x = 45
x = 5

Number of boys 4×5 = 20
Number of Girls 5×5 = 25
Total marks of boys 75×20 = 1500
Total marks of girls 25×82= 2050
Total marks of girls and boys = 1500+2050 = 3550

Average of marks of class $= \frac{3550}{45} = 78.88$, = 78.9

35. The average test score of 18 boys in a class was 15, while the over all average of all the 25 students of the class was 16.12. What was the average score of the girls?

(a) 18.5 (b) 19.5
(c) 19 (d) 18.8

RRB Group-D – 01/11/2018 (Shift-II)

Ans : (c) Number of girls = 25 – 18 = 7
Hence the average test score of a girls

$$= \frac{25 \times 16.12 - 18 \times 15}{7}$$

$$= \frac{403 - 270}{7} = \frac{133}{7} = 19$$

36. The average marks of students in 5 subject is 75. Average of his first two subjects is 65 and the average of his last two subjects is 85. How much did he score in the third subject?

(a) 80 marks (b) 65 marks
(c) 75 marks (d) 70 marks

RRB Group-D – 12/12/2018 (Shift-I)

Ans. (c) Average marks of obtained by the student in 5 subjects = 75

Total marks obtained by the students = 75 × 5 = 375

$\because$ Average of first 2 subjects = 65

Sum of first 2 subjects = 65 × 2 = 130

$\because$ Average of last 2 subjects = 85

Sum of last two subjects = 85 × 2 = 170
Obtained marks in third subject =
sum of all five subjects − (sum of first two subjects+ sum of last two subjects)
= 375 – (130 + 170) = 375 – 300 = 75
Hence obtained marks in third subject is = 75

37. In a class of 50 students, the ratio of boys and girls is 2:3. The average marks of the boys is 60 and the average marks of the girls is 70. What is the average of the class?

(a) 65 (b) 66
(c) 67 (d) 64

RRB Group-D – 12/11/2018 (Shift-II)

Ans : (b) Given–
Ratio of girls and boys is 2 : 3

Number of boys $= \frac{50 \times 2}{5} = 20$

Number of girls $= \frac{50 \times 3}{5} = 30$

Average marks of whole class $= \frac{60 \times 20 + 30 \times 70}{50}$

$$= \frac{1200 + 2100}{50} = \frac{3300}{50} = 66$$

38. A group of 19 students took a test, one more students took test and joined the group. By including his marks, the average is raisedby 1.5 marks. Without including him this student obtained ____ more marks than the average?

(a) 25 (b) 30
(c) 24 (d) 28.5

RRB Group-D – 12/10/2018 (Shift-II)

Ans : (b) Let the Average marks of 19 students = x

$$\frac{student_1 + student_2 + ---- + student_{19}}{19} = x$$

$student_1 + student_2 + ----- + student_{19} = 19x$ ---- (i)

After including new student–

$$\frac{student_1 + student_2 + ---- + student_{20}}{20} = x + 1.5$$

$student_1 + student_2 + ----- + student_{20} = 20x + 30$ --- (ii)

by subtracting equation (i) from equation (ii)

$student_{20} = 20x + 30 - 19x = x + 30$

It is clear that the student included in the group has scored 30 marks more than the average-

39. The average marks obtained by Suvir in 15 exams is 29. Ruchira has maintained the average of 27 but till now she has taken only 11 tests. How much Ruchira has to perform in the rest test to match the average of Suvir performance?

(a) 35 (b) 34.5
(c) 36 (d) 35.5

RRB Group-D – 04/10/2018 (Shift-I)

Ans. (b) Given-

Average marks of suveer in 15 examinations = 29

Total marks of suveer = 29 × 15 = 435

Average marks of Ruchira in 11 examinations = 27

Total marks of Ruchira = 27 × 11 = 297

Total required marks of Ruchira in remaining 4 examinations = 435−297 = 138

Hence, Required average marks of Ruchira $= \frac{138}{4} = 34.5$

40. A group of seven students took an exam. After exam one more students joined the group. After including the new students the average marks is raised by 2. In comparison with the average marks of initial seven students, how much more marks did the new student obtain?

(a) 16 (b) 18
(c) 14 (d) 20

RRB Group-D – 01/10/2018 (Shift-II)

Ans. (a) : Let Average marks of 7 students = x

∴ Total obtained marks by 7 students = 7x

Again let the marks of new student = y

According to question,

$$\frac{7x + y}{8} = (x + 2)$$

7x + y = 8x + 16

y = x + 16

Hence, student scored 16 marks more than the average marks.

41. In a class the average marks obtained by 35 students is 63. If two more students whose average is 85.5 is added, then the new average of the class is?

(a) 64.20 (b) 67.90
(c) 63.62 (d) 65.35

RRB NTPC 04.04.2016 Shift : 2

Ans : (a) Total sum of marks of 35 students

= 35×63=2205

from question-

New Average of class $= \frac{2205 + 2 \times 85.5}{35 + 2}$

$$= \frac{2205 + 171}{37} = \frac{2376}{37} = 64.21 \cong 64.20$$

42. The average marks obtained by James in maths, science and history is 89. If the marks of Language is also added then the averages decreases to 88.25. Find out the marks obtained by him in the language.

(a) 90 (b) 82
(c) 86 (d) 83

RRB NTPC 28.03.2016 Shift : 3

Ans : (c) maks obtained by him in the language

= 4 × 88.25 − 3 × 89

= 353 − 267 = 86

Type - 7

43. In a class of 10 students, the average age was 16 years. When two students dropped the class then the average age of the rest students was 16.25 years. What was the total age of the dropped students?

(a) 32 years (b) 30 years
(c) 34 years (d) 28 years

RRB RPF Constable – 20/01/2019 (Shift-III)

Ans : (b) Total average age of 10 students

= 16 × 10 = 160

Total average age of remaining 8 students = 8 × 16.25

= 130.00

Total age of dropout students = 160–130 = 30 Years

44. In a group of 5 people the ratio of the average age of first three and the last two is 9:7. If the difference of their age is 12, then what will be the average age of all the five?

(a) 46.8 (b) 49.2
(c) 48.4 (d) 64.8

RRB Group-D – 03/10/2018 (Shift-II)

Ans : (b) Average age of group of 3 people = 9 x years
Average age of group of 2 people = 7x years
Average age difference of both group = 12
$\therefore$ 9x – 7x = 12
2x = 12
$\boxed{x = 6}$
Average age of 3 people = 9 × 6 = 54 years
Average age of 2 people = 7 × 6 = 42 years
Total age of 3 people = 3 × 54 = 162 years
Total age of 2 people = 2 × 42 = 84 years
Total age of 5 people = (162+84) = 246 years
Average age of 5 people $= \frac{246}{5} = 49.2$ years

45. The average of 40 students age is 30 years, average of 25 students age is 36 years. Find the average age of the rest of the students?
(a) 20 (b) 15
(c) 25 (d) 18
RRB NTPC 05.04.2016 Shift : 2

Ans : (a) Average age of remaing students
$$= \frac{40\times30-25\times36}{40-25}$$
$$= \frac{1200-900}{15}$$
$$= \frac{300}{15} = 20 \text{ years}$$

46. If the average age of 40 students of class first is 10 years and the average age of 30 students in class IInd is 12 years. Find the average age of all students (in years).
(a) 11 (b) 10.54
(c) 10.58 (d) 10.85
RRB NTPC 29.04.2016 Shift : 2

Ans : (d) Total age of 40 students = 10×40 = 400 years
Total age of 30 students = 12 × 30 = 360 years
Average age of all students $= \frac{400+360}{70}$
$= \frac{760}{70} = 10.85$ years

Type - 7

47. Average of 81 results is 54. If the average of first 59 results is 52 and last 21 results is 60, then calculate the 60th result?
(a) 52 (b) 60
(c) 46 (d) 62
RRB Group-D – 11/12/2018 (Shift-I)

Ans. (c) : Grand Total of 81 results = 81×54= 4374
Grand Total of first 59 results = 59 × 52 = 3068
Grand Total of last 21 results = 21 × 60 = 1260
60th result = 4374 – 3068 – 1260 = 46

48. Average of 11 results is 50. The average of first 6 results is 49 and last 6 results average is 52. Then what will be the value of the 6th result?
(a) 48 (b) 51
(c) 56 (d) 49
RRB NTPC 02.04.2016 Shift : 3

Ans : (c) 6th result = grand total of first 6 results + grand total of last 6 results - grand total of 11 results.
$$= 6\times49+6\times52-11\times50$$
$$= 294+312-550$$
$$= 606-550 = 56$$

49. The average of 45 results is 23. The average of first 22 is 18 and last 22 is 21. What is the value of 23 result?
(a) 172 (b) 190
(c) 177 (d) 187
RRB NTPC 18.04.2016 Shift : 3

Ans : (c) Total value of 45 results = 45 × 23 = 1035
Average of first 22 results = 22 × 18 = 396
Averge of last 22 results = 22 × 21 = 462
value of 23rd results = 1035 – (396+462)
= 1035–858 = 177

Type - 5

50. A person received 4 packets in January, whose average weight was 300g and in February 8 packets were received whose average weight was 400g. What will be the average weight (g) in received by that person in the two months?
(a) 350g (b) 366.67g
(c) 412.67g (d) 376.67g
RRB RPF SI – 10/01/2019 (Shift-I)

Ans : (b)
In January, total weight of 4 packets = 4×300 = 1200 g
In February, total weight of 8 packets = 8×400 =3200g
Average weight of total packets
$$= \frac{\text{sum of total weight}}{\text{Number of total packets}}$$
$$= \frac{1200+3200}{12} = \frac{4400}{12} = \frac{1100}{3}$$
= 366.67 g

51. The average weight of 25 items is 50 kg. If any other item X weight included, then average weight is increased by 500 g. What is the weight of item X ?
(a) 28 kg (b) 36 kg
(c) 82 kg (d) 63 kg
RRB Group-D – 25/09/2018 (Shift-II)

Ans : (d) Total weight of 25 goods = 25 × 50
= 1250 kg
Total average weight of including the weight of the new object X
= (50 + .5) kg = 50.5 kg
Hence total weight of total 26 goods = 50.5 × 26
= 1313 kg
Weight of X = 1313 – 1250 = 63 kg

52. The mean weight of six kids is 17.5 kg. If the individual weights of 5 of those kids are 14, 19, 23, 21 and 13 kg, respectively, find the weight of the sixth kid.

(a) 17 kg (b) 15 kg
(c) 16 kg (d) 18 kg

RRB Group-D – 05/11/2018 (Shift-III)

Ans. (b) : Let the weight of 6th child is x kg

$\therefore 17.5 = \frac{14+19+23+21+13+x}{6}$

105.0 = 90 + x

x = 15

Hence weight of 6th child = 15 kg

53. The average weight of 42 boys of a class is 41 kg. A boy of 39 kg weight joins the class. A boy already present in the class, whose weight was counted as 34 kg in place of 43 kg. What will be the new average?

(a) 39.81 (b) 40.74
(c) 41.16 (d) 40.92

RRB NTPC 26.04.2016 Shift : 1

Ans : (c) Total weight of 42 boys = 42 × 41 = 1722

On inclusion of 39 kg boy

New weight = 1722 + 39 = 1761

According to question,

New Average = $\frac{1761+43-34}{43} = \frac{1770}{43} = 41.16$

54. Average of 5 person weight is 76 kg. in these four persons weight is 72, 74, 75 and 81 kg. 5th persons weight in kg will be.

(a) 77 (b) 78
(c) 79 (d) 80

RRB NTPC 26.04.2016 Shift : 3

Ans : (b) Total weight of 5 people $= 76 \times 5 = 380$

weight of 4 people $= 72 + 74 + 75 + 81$

$= 302$

weight of 5th person $= 380 - 302 = 78$ kg

55. In a class the average of 30 student age is 25 kg. If the class teacher's weight is also included in this, then average weight is increased by two kg. What is the weight of the class teacher (in kg) ?

(a) 86 (b) 84
(c) 81 (d) 87

RRB NTPC 27.04.2016 Shift : 1

Ans : (d) Weight of teacher in the class = 31 × 27 – 30 × 25

= 837 – 750 = 87 kg

56. The average weight of 29 students of any class is 48. If the class teacher weight is added then the average weight is increased by 1/2 kg. Class teacher weight (in kg) is?

(a) 63 (b) 65
(c) 60 (d) 68

RRB NTPC 29.04.2016 Shift : 3

Ans. (a) : Total weight of 29 students = 29 × 48 = 1392 when the weight of the class teacher is added.

Then total weight = 30 × 48.5 = 1455 kg

∴ weight of class-teacher = 1455 – 1392 = 63 kg

57. The average weight of 30 students of a class is 15 kg. If the weight of the class teacher is included in it then the average weight is increased by 1 kg. What is the weight of the class teacher (in kg).

(a) 46 (b) 45
(c) 16 (d) 92

RRB NTPC 30.04.2016 Shift : 1

Ans : (a) Total weight of 30 students in the class $= 30 \times 15 = 450$

Total weight if class teacher is also included

$= 31 \times 16 = 496$

Hence the weight of class teacher $= 496 - 450 = 46$ kg

Type - 6

58. The average score of a group of cricketers was 42. A new player joins and score 250% of the average of the group. As a result the overall average is increased by 30% what was the numbers of cricketers in the group before, the new player joined?

(a) 3 (b) 5
(c) 6 (d) 4

RRB RPF Constable – 22/01/2019 (Shift-II)

Ans : (d) Let the Number of cricketers in the group =x, and average = 42

the, total score = 42x

Number of cricketers after including one player in the group = x + 1

New average $= \frac{42 \times 130}{100}$

= 54.6

Score of new player $= \frac{42 \times 250}{100} = 105$

Now total score = (105 + 42x)

Hence sum = average × number

105 + 42x = 54.6 (x + 1)

105 – 54.6 = 54.6x – 42x

50.4 = 12.6x

$x = \frac{50.4}{12.6} = 4$

Therefore number of cricketers in first group (x) = 4

59. After 11 innings, the average score of a batsman is 52 per innings. After 13 innings, the average is raised to 54. If the batsman made 16 runs more than the last inning in the 13 inning then how much did he make in 12th inning?

(a) 54 (b) 57
(c) 56 (d) 55

RRB Group-D – 25/09/2018 (Shift-III)

Ans. (b) : Total score after 11 innings = 52×11 = 572

total score after 13 innings = 54×13 = 702

12th inning+ score of 13th inning = score of 13 innings – score of 11 innings = 702 − 572 = 130

According to question-

Let the Run in 12th innings = x

Run in 13th innings = x+16

then, $x + (x+16) = 130$

$2x = 130 - 16$

$2x = 114$

$\boxed{x = 57}$

60. The average score of 12 members of a group was 8. While another group of n members had an average score of 10. If the joint average was 9.2 find the value of n.

(a) 18 (b) 24

(c) 16 (d) 30

RRB Group-D – 24/10/2018 (Shift-II)

Ans. (a) : Sum of group of 12 members = 12×8= 96

Sum of the group of n members = 10×n = 10n

∵ Combined average = 9.2

∴ According to question-

$$9.2 = \frac{96+10n}{(12+n)}$$

$$(12+n) = \frac{960+100n}{92}$$

$$1104+92n = 960+100n$$

$$144 = 8n$$

$$n = \frac{144}{8} = 18$$

$$\therefore n = 18$$

61. The average run scored by a batsman is 125 in 5 matches. In the first two matches is average score was 150. The average of runs scored in the last two matches is 110. How many runs that batsman scored in the third match?

(a) 115 run (b) 125 run

(c) 105 run (d) 95 run

RRB Group-D – 03/12/2018 (Shift-III)

Ans. (c) : Total score make in 5 matches = 125×5= 625

Total score make in first 2 matches = 150 × 2 = 300

Total score make in last 2 matches = 110 × 2 = 220

∴ Total runs scored in the third match

= 625–(300+220)

= 625 – 520

= 105 run

62. After 12 innings, a batman per inning average score was 55. After 14 innings his average score reached to 60. If the batsman scored 20 runs more in the 14th innings than the last innings, then how much did he score in the 13th innings?

(a) 90 (b) 85

(c) 80 (d) 75

RRB Paramedical Exam – 21/07/2018 (Shift-I)

Ans : (c) Average score of 12 innings = 55

∴ Total score = 55×12 = 660

Average score of 14 innings = 60

∴ Total score = 14×60 = 840

Total score of 12th and 14th innings = 840 – 660 = 180

Let the score of 13th inning is x. then score of 12th inning = x + 20

According to question-

∴ x + 20 + x =180

2x = 160 ⇒ x = 80

63. In a football tournament Real madrid did 33 goals and 22 goals were done against it. Find the average number of goals done by each player of the real madrid?

(a) 1 (b) 2

(c) 3 (d) 0

RRB NTPC 30.04.2016 Shift : 2

Ans : (c) Real Madrid average number of goals scored for each palyer

$$= \frac{\text{Number of goals in real madrid}}{\text{number of players of football}} = \frac{33}{11} = 3$$

64. After 10 innings the average score per innings of a batsman was 52. After 12 innings the average score to 54. If the batsman had scored 16 more runs in the 12th innings than in the previous one, how many runs did he score in the 11th innings?

(a) 55 (b) 56

(c) 54 (d) 53

RRB ALP & Tec. (14-08-18 Shift-I)

Ans : (b) According to question,

Total score of 10 innings = 10 × 52 = 520

Total score of 12 innings = 12 × 54 = 648

Let, makes x run in 11th innings

According to question,

$$x + x + 16 = 648 - 520$$

$$2x = 128 - 16$$

$$2x = 112$$

$$x = 56 \text{ run}$$

Hence the bats man has scored 56 runs in 11th innings.

Type - 7

65. The sales quanity in a shop is recorded for four different quarters. Following data is-

Quarter	Selling Amount
Quarter 1	200
Quarter 2	100
Quarter 3	350
Quarter 4	550

per quarter average selling is?

(a) 300 (b) 500

(c) 250 (d) 350

RRB RPF SI – 12/01/2019 (Shift-III)

Ans : (a) Average sales per quarter

$$= \frac{(200+100+350+550)}{4}$$

$$= \frac{1200}{4} = 300$$

66. Based on the following table, what is the average number of screws manufactured in the unit in given 6 months?

Months	No. of screws manufactured
January	200
February	300
March	250
April	250
May	250
June	250

(a) 300 (b) 200
(c) 250 (d) 150

RRB Group-D – 28/09/2018 (Shift-II)

Ans. (c) : Average = $\frac{\text{sum of terms}}{\text{number of terms}}$

Average of the screw

$$= \frac{200+300+250+250+250+250}{6}$$

$$= \frac{1500}{6} = 250$$

67. A private swimming pool offers different time slots for its users, and the following table represents the number of pool visitors of the week?

Days	No. of users of swimming pool
Monday	8
Tuesday	4
Wednesday	15
Thursday	15
Friday	20
Saturday	25
Sunday	25

On average how many visitors visited on a day of that week?

(a) 12 (b) 16
(c) 14 (d) 15

RRB Group-D – 15/10/2018 (Shift-I)

Ans : (b) Number of visitors in the whole week =
8 + 4 + 15 + 15 +20 +25 + 25 = 112

average = $\frac{112}{7} = 16$

Hence, an average of 16 visitors visited throughout the week.

68. Two football teams, Team A and Team B played in a tournament and following are the goals done in six matches.

Play	Goal by Team A	Goal by Team B
Kesue 1	2	3
Kesue 2	1	0
Kesue 3	0	1
Kesue 4	4	5
Kesue 5	3	2
Kesue 6	2	1

Based on average targets, what conclusion we can derive?

(a) The average score of Team A and B is same.
(b) Team A's average score is comparatively more than B.
(c) Team A is similar to B.
(d) Team B's average score is comparatively more than A.

RRB Group-D – 26/09/2018 (Shift-III)

Ans : (a) Average score of Team A

$$= \frac{2+1+0+4+3+2}{6} = \frac{12}{6} = 2$$

Average score of Team B

$$= \frac{3+0+1+5+2+1}{6} = \frac{12}{6} = 2$$

69. What is the difference in average price per kg (in ₹) for vegetables, three months between two years, according to the data given in the following table?

Months	Per Kg Price-year 1	Per Kg Price-year 2
January	40	35
February	40	50
March	40	35

(a) There is no difference in the average price.
(b) We can't compare values.
(c) The average price of vegetables in year 1 is less than year 2.
(d) The average price of vegetables in year 2 is less than year 1.

RRB Group-D – 27/09/2018 (Shift-I)

Ans. (a)

Average price of first years $= \frac{40+40+40}{3} = 40$

Average price of second years $= \frac{35+50+35}{3} = 40$

Difference = 40 − 40 = 0

Hence, there is no difference in the average price.

70. The number of people going through the way of shop is recorded for four different quarters. The following data is given for this. What is the average number of people going through the shop way in all the quarters?

Quarters	No. of People
Quarter1	2,000
Quarter2	1,000
Quarter3	3,500
Quarter4	5,500

(a) 12,000 (b) 4,000
(c) 10,000 (d) 3,000

RRB Group-D – 22/10/2018 (Shift-III)